Sensor Fusion for Human Activity Recognition: A Multimodal Evaluation on the MHEALTH Dataset

Md Soad Solaiman

Abstract

This study tested whether combining wearable sensor measurements improved activity classification. The adapted dataset contained 343,195 rows from ten participants and 12 activities. Short internal gaps were interpolated. The recordings were then divided into overlapping windows. This produced 4,581 usable windows. Random Forest models were tested on one unseen participant at a time.

The models used acceleration and angular velocity from three body locations. Feature fusion combined their extracted features. Decision fusion averaged their class probabilities. The combined accelerometer baseline achieved the highest pooled macro F1 (96.91%). Feature fusion reached 96.51% and decision fusion 96.45%. Although fusion improved results for some participants, it reduced the overall pooled score. Under this design, adding gyroscopes brought no clear advantage.

Task 1 Data exploration

1.1 Dataset and sensor structure

Table 1 summarises the recorded and selected channels. Each participant had one log file. Each row contained 23 sensor measurements and one activity label. The original MHEALTH documentation specifies a sampling rate of 50 Hz (Uddin and Soylu, 2021). This rate was used to convert sample counts into seconds.

Table 1: Recorded channels and inputs selected for modelling

Device location	Recorded channels	Selected channels
Chest	3 acceleration and 2 ECG	3 acceleration
Left ankle	3 acceleration, 3 gyroscope and 3 magnetometer	3 acceleration and 3 gyroscope
Right wrist or lower arm	3 acceleration, 3 gyroscope and 3 magnetometer	3 acceleration and 3 gyroscope

ECG means electrocardiogram. A source means a physical device. A modality means a type of measurement. The selected inputs covered three devices and two modalities. ECG and magnetometer channels were retained during the initial quality analysis but were not used by the models.

1.2 Activity coverage and movement patterns

The files were combined into one table. Participant IDs and original row positions were preserved. A new segment began when the activity label changed. This kept each continuous labelled segment identifiable. All ten participants had all 12 activity labels.

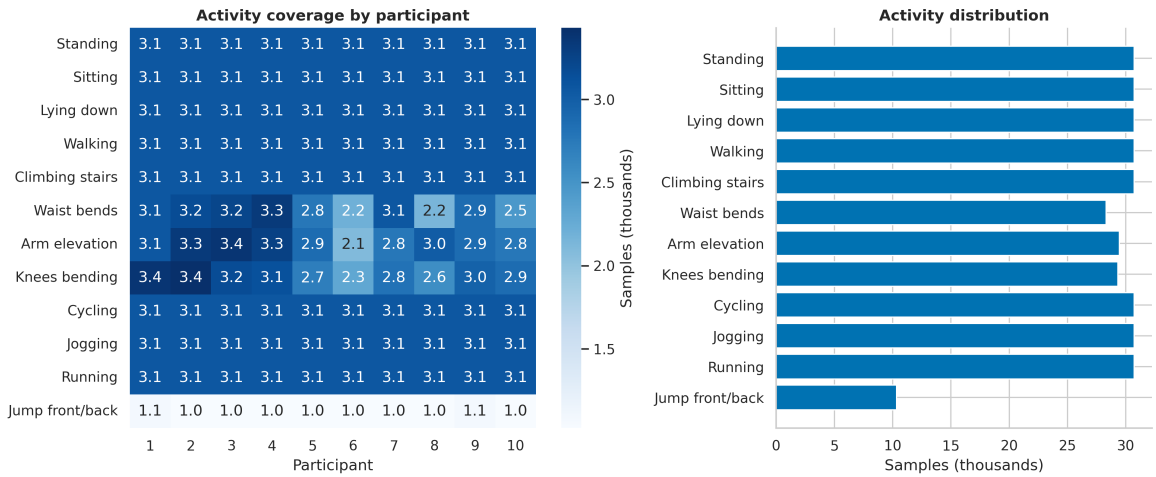


Figure 1: Activity coverages. Counts are shown in thousands

Figure 1 shows uneven activity coverage. Jumping had the fewest samples. This means accuracy alone could hide weak performance on smaller classes. Macro metrics were therefore included in the evaluation.



Figure 2: First five seconds of four contrasting activities for participants 1, 6 and 10. The dashed line marks gravity. Missing observations remain gaps

Figure 2 shows raw chest acceleration magnitude for standing, walking, jogging and running. Standing stays near gravity, approximately 9.8 m/s^2. Dynamic activities show larger changes from movement. These patterns motivate the level and spread features in Task 2. Differences between participants also motivate the LOSO (Leave-One-Subject-Out) design in Task 3.

Task 2 Missing data and preprocessing

2.1 Missingness and gap handling

There were 615,648 missing sensor values, or 7.80% of sensor cells. Only 15.13% of rows had all 23 channels present. Deleting every incomplete row would therefore remove most observations and disrupt the sampling sequence.

Missingness differed by channel. It ranged from 1.13% in wrist gyroscope Y to 14.67% in ankle magnetometer Z. Figure 3 shows a similar channel pattern across participants. Overall percentages (right of figure 3) were also similar across activities. These patterns do not establish why values were missing.

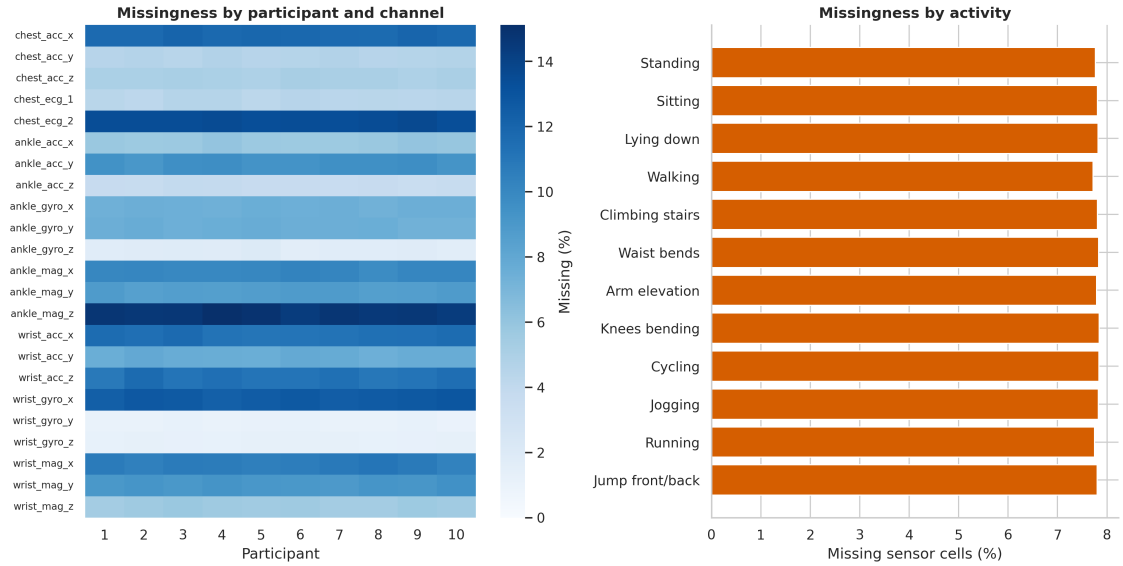


Figure 3: Missing sensor values by channel and participant, and by activity.

Missing runs were counted separately within each participant and activity segment. Here, a run means values missing in a row. The longest run contained seven samples. It means seven missing values in a row. Only 12 runs exceeded five samples. A maximum gap of five samples was therefore chosen for interpolation. At the documented sampling rate, this equals 0.10 seconds.

Linear interpolation filled only complete internal gaps of five samples or fewer. Leading gaps, trailing gaps and longer gaps were left unchanged. After interpolation, it filled 615,103 cells and left 545 missing.

2.2 Windows and feature extraction

Each segment was divided into 128-sample windows. This corresponds to 2.56 seconds at 50 Hz (Rehman et al., 2024). The stride was 64 samples, giving 50% overlap. A window could not contain two activities or two participants. Half-window overlap reduces sensitivity to the starting position. It also creates dependent windows, which is why whole participants must remain together during evaluation.

A window was rejected if any channel originally had more than 20% missing values. It was also rejected if any value remained missing after interpolation.

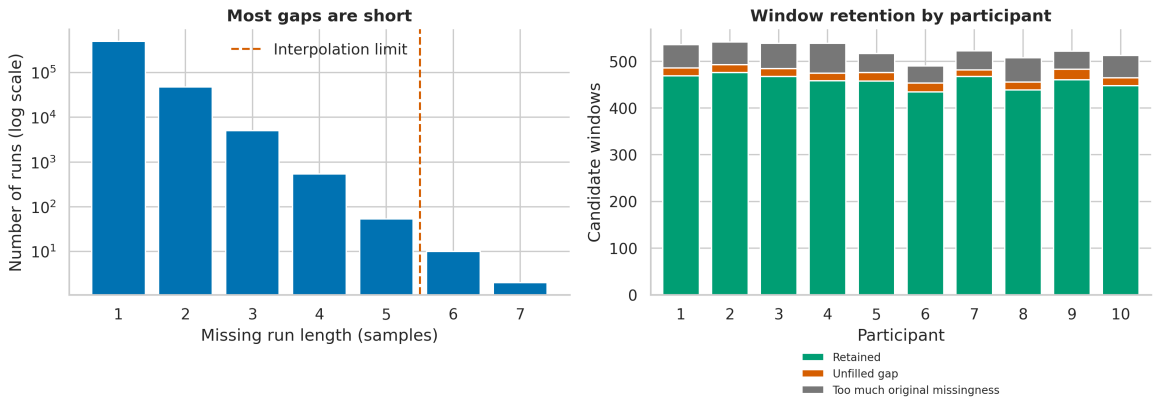


Figure 4: Missing run lengths and the effect of the window eligibility rules.

There were 5,229 candidate windows. Of these, 4,581 were retained, or 87.61%. A further 474 failed the original-missingness rule. Another 174 contained an unfilled gap. The unused segment tails contained 859 sample positions. These counts are not additive raw-data losses because windows overlap. Every participant still had all 12 activities.

2.3 Preventing leakage

Leakage occurs when test information reaches the model during training, inflating results (Kapoor and Narayanan, 2023). The following steps prevented it:

- Participant IDs, labels and sample positions were never used as model inputs.
- Each fold tested on one whole participant.
- All of that participant's overlapping windows stayed in the test set.
- No scaling, feature selection or population-based imputation was fitted across participants.
- Interpolation used only data from the same participant and activity segment.

Task 3 Multimodal fusion and modelling

3.1 Input selection and baselines

Different sensing modalities capture different aspects of activity (Bian et al., 2022). Acceleration was selected from the chest, ankle and wrist. Angular velocity was selected from the ankle and wrist. This gave 15 channels and 90 features. Chest measurements described trunk movement. Ankle and wrist measurements added lower-limb and arm movement (Aguileta et al., 2019).

Table 2: Experiments and predictive inputs

Experiment	Features	Modalities	Forests
Chest accelerometer	18	1	1
Ankle accelerometer	18	1	1
Wrist accelerometer	18	1	1
Combined accelerometers	54	1	1
Combined gyroscopes	36	1	1
Feature fusion	90	2	1
Decision fusion	90 in total	2	2

Table 2 follows a progressive comparison. First, each body location was tested alone. Next, the three accelerometers were combined. Finally, gyroscope information was added through feature and decision fusion. The individual accelerometer models provided source-level references. The combined accelerometer model used several locations but only one modality. The combined gyroscope model was also unimodal.

3.2 Classifier and training procedure

Random Forest combines decision trees trained with randomised samples and feature choices (Thakur and Biswas, 2023). It was chosen to model nonlinear relationships in these tabular features. It required no scaling. Every model used 200 trees. Feature sampling used the square-root rule. The minimum leaf size was two, and tree depth was unrestricted. Class weights were balanced using training labels only. The random seed was 42. One worker was used for consistent timing. These settings were fixed before testing. No hyperparameter search or inner validation was performed. The settings are not claimed to be optimal.

Each fold trained on nine participants and tested on the remaining participant (Bragança et al., 2022). All ten folds were completed. There were 60 forest fits. Decision fusion reused the modality models. All comparisons used the same test windows.

Task 4 Evaluation and critical discussion

4.1 Overall performance

All saved predictions were checked before interpretation. Each model had one held-out prediction for every eligible window. Labels, participants and class order matched. Probability rows summed to one. Recomputed fold and pooled metrics matched the Task 3 results. Table 3 reports pooled scores.

Table 3: Pooled held-out performance in percent

Model	Accuracy	Macro precision	Macro recall	Macro F1
Combined accelerometers	96.81	97.22	96.79	96.91
Feature fusion	96.44	96.85	96.34	96.51
Decision fusion	96.44	96.87	96.21	96.45
Ankle accelerometer	85.96	87.81	86.91	87.09
Chest accelerometer	82.71	84.32	83.55	83.40
Combined gyroscopes	78.63	78.42	74.86	75.28
Wrist accelerometer	73.96	75.29	73.39	73.74

Precision, recall and F1 used all 12 labels and explicit zero-division handling. Macro averaging gave every activity equal weight. The combined accelerometers had the highest macro F1. The strongest baseline was identified from these results, so its ranking was descriptive rather than an independent selection result.

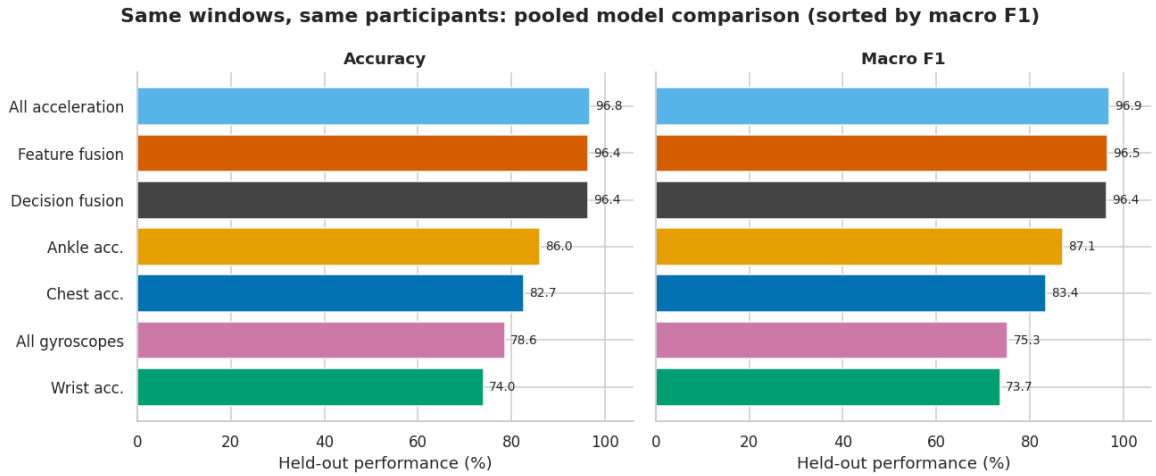


Figure 5: Pooled accuracy and macro F1 on the same held-out windows.

Feature fusion reduced macro F1 by 0.40 in percentages relative to combined accelerometers. Decision fusion reduced it by 0.47 in percentages. Both reached 96.44 in percentages accuracy, below the baseline's 96.81 in percentages. Both fusion methods still outperformed the individual accelerometer models and the combined gyroscopes. Adding locations helped the acceleration model here. Adding gyroscopes did not improve its pooled results. But the question is whether the F1 results are almost the same for every person on every test, or whether they vary. If they vary, how much do they vary, and how confident can we be in the pooled results?

4.2 Participant variation and uncertainty

The performance variations and the bootstrap are answering the questions. Figure 6 shows the strongest baseline and both fusion models. Feature fusion improved macro F1 for five participants, worsened it for three and tied for two. Decision fusion improved six and worsened four. Improvements in several people were offset by larger losses elsewhere.

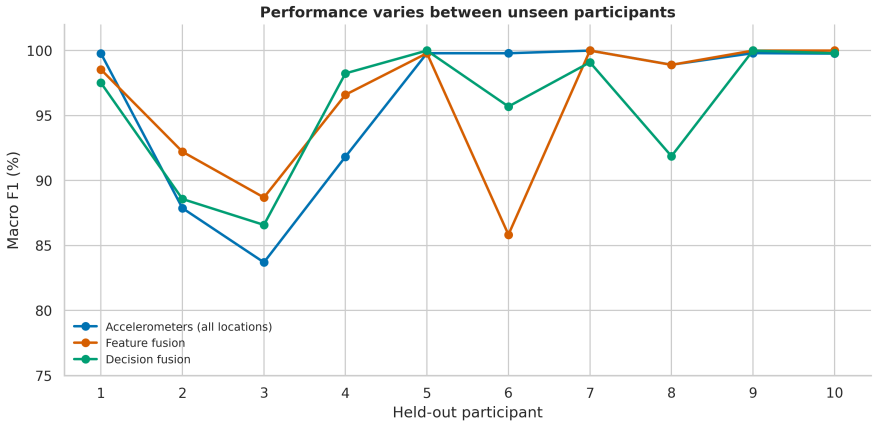


Figure 6: Macro F1 for each held-out participant. The vertical axis starts at 75%.

Table 4: Participant macro F1 summaries in percent. SD is the sample standard deviation.

Model	Mean	SD	Minimum	Maximum
Combined accelerometers	96.13	6.06	83.70	100.00
Feature fusion	96.06	5.26	85.83	100.00
Decision fusion	95.74	4.98	86.58	100.00

Table 4 summarises the participant results. The baseline performed worst on participant 3, at 83.70% macro F1. Feature fusion performed worst on participant 6, at 85.83%. Decision fusion performed worst on participant 3, at 86.58%. Decision fusion had the smaller spread among the fusion methods. This was a descriptive finding from ten people. The causes of participant differences were not tested.

4.3 Bootstrap uncertainty

Uncertainty was assessed with 2,000 bootstrap resamples and seed 42. Whole participants were resampled, including all their windows. The same draws were used for every model. Pooled metrics were recalculated for each draw. This respected participant clustering instead of treating overlapping windows as independent.

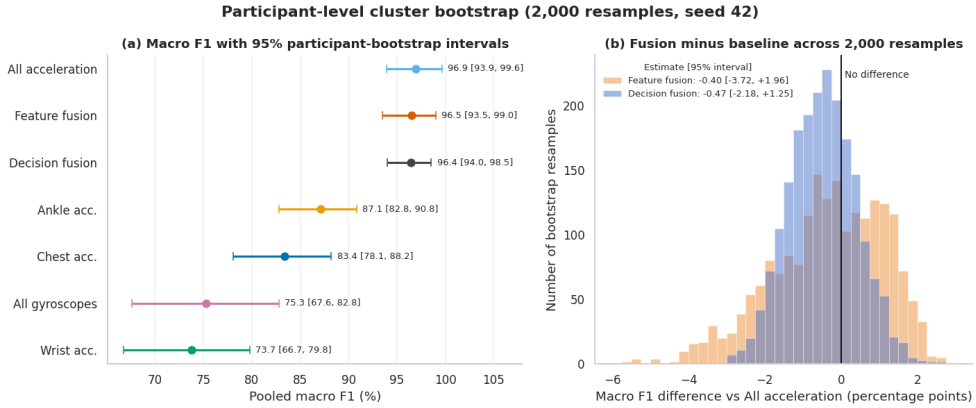


Figure 7: Bootstrap clustering for the 95% confidence level of Macro F1.

The left confidence plot of the figures shows that, based on the available data, we are at least 95% confident that the overall performance lies within the 95% confidence interval. Also, the histogram shows that both fusion methods carry no clear difference. But we still don't know which activity is actually causing the loss of accuracy and macro F1.

4.4 Activity results and confusion patterns

From these confusion matrix patterns, the underlying cause can be identified.

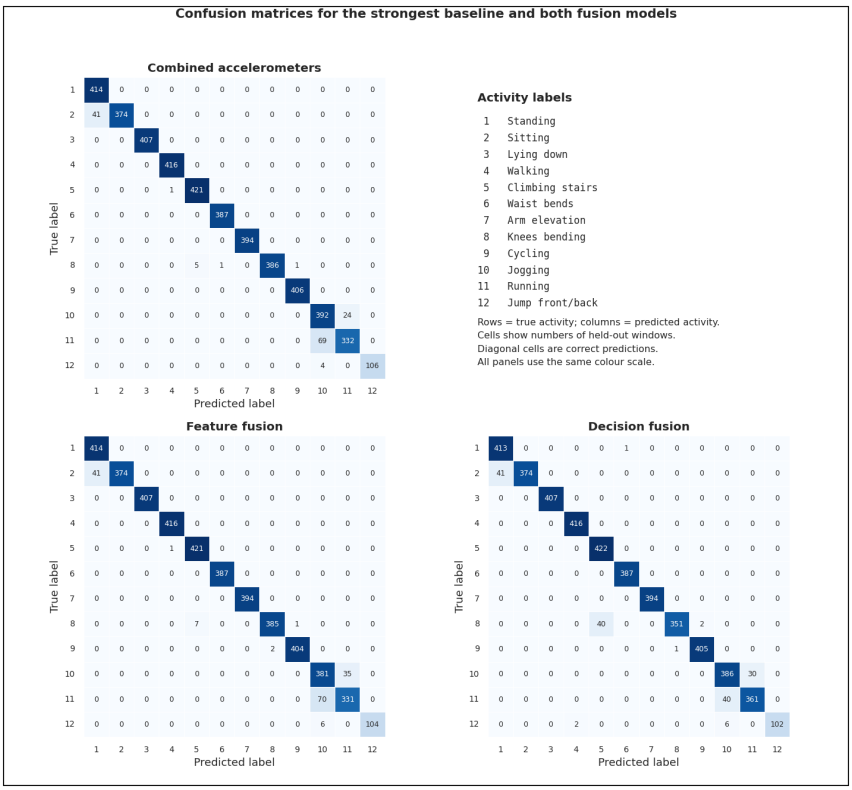


Figure 8: Row-normalised confusion matrices. Numbers are percentages of each true activity.

Figure 8 shows that most activities were classified almost perfectly. Four confusion patterns remained.

Running and jogging were confused most, mainly by feature fusion (70 errors) versus decision fusion; the reverse error was rarer, both sharing gait cycles differing in speed. Sitting and standing were each mislabelled 41 times, as no thigh sensor captured their key difference. Decision fusion confused knees-bending with climbing stairs 40 times, since both bend the legs. Accelerometer and feature fusion performed best, though accelerometer slightly led. Now the question arises why this is happening despite the fusion. What is the contribution of fusing the gyroscope, then?

4.5 What feature importance shows

The answer can be found from this LOSO (Leave-One-Subject-Out) analysis. Figure 9 shows that accelerometers dominated the feature-fusion models, contributing 88.59% of normalised impurity importance compared with only 11.41% for gyroscopes. The ankle, chest and wrist accelerometers accounted for approximately 36%, 28% and 24% respectively. This connects the source comparison in Task 3 to the strong spatial accelerometer baseline in Table 3.

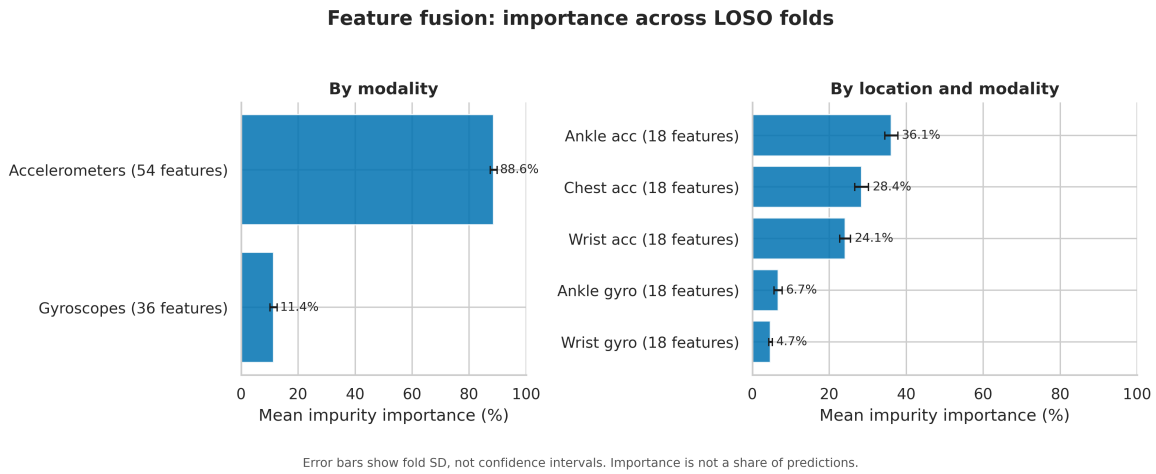


Figure 9: Mean normalised impurity importance across ten feature-fusion forests. Whiskers show fold standard deviations, not confidence intervals. Feature counts are shown beside each group.

Importance measures weighted reductions in training impurity. It is not a percentage of predictions or decisions. Correlated features can share importance, and group totals depend partly on feature counts: 54 acceleration features versus 36 gyroscope features. The result shows lower relative use of gyroscopes.

After fusing, does it take more or less time to predict? What about the timing?

4.6 Computational implications

Table 5: Measured computation in seconds across all ten folds

Model	Training (sec)	Prediction (sec)	Total (sec)
Chest accelerometer	21.000	0.167	21.167
Ankle accelerometer	20.567	0.166	20.733
Wrist accelerometer	20.926	0.196	21.122
Combined accelerometers	36.223	0.179	36.403
Combined gyroscopes	30.585	0.177	30.763
Feature fusion	36.248	0.151	36.399
Decision fusion	66.809	0.357	67.165

Table 5 shows training time far exceeding prediction time (under half a second) across all ten folds for every model. Decision fusion was slowest at 67.2 seconds, since it trains separate accelerometer and gyroscope forests whose times roughly sum. Feature fusion took 36.4 seconds, matching the accelerometer-only model, showing the 36 added gyroscope features barely affected training speed.

The two fusion strategies trade off differently: feature fusion combines sensors but risks redundant information, while decision fusion is modular yet weights both sensor models equally, letting the weaker gyroscope model skew results. Neither was tested with a missing sensor. Timings are hardware-dependent and exclude loading, extraction, memory, battery, and model-size costs; more sensors also mean more real-world cost and maintenance.

4.7 Ethical considerations

Wearable records can reveal movement routines and sensitive health information, and the raw archive contains ECG data even though unused here (Sui et al., 2023). Consent should explain the sensors, purpose, retention period and secondary uses, with clear withdrawal rights.

Deployment would need secure storage and transmission, restricted access, clear ownership, and retention/deletion rules, since a breach could expose private routines. Reusing data for workplace surveillance or insurance would need separate justification, and users should be able to challenge

incorrect predictions and get human review.

Demographic information was unavailable, so fairness across age, sex, disability, health or cultural group could not be assessed; comparing scores does not establish fairness. These results do not justify clinical, employment, insurance or safety-critical use.

4.8 Limitations and improvements

Small, structured dataset. The data came from ten people performing 12 set activities, and no external dataset was tested. Participants followed a fixed routine (Uddin and Soylu, 2021). Future work should test more varied people and natural daily-life recordings.

Unknown cause of missing data. It was not explained how or why values were removed. The gap filling therefore does not show robustness to real sensor failures. Future work should test realistic sensor dropouts.

Offline preprocessing. The pipeline knew where each activity started and ended, and interpolation used future values. Neither is available in real time. Windows were also removed because of channels the models did not use. Future work should test continuous data and apply the window rule to the selected sensors only.

Simple features and one model. The features ignored the order of values within each window. Only Random Forest with fixed settings was tested. Future work could try tuned or sequence models, tuned on training participants only.

Equal fusion weights. The two modality models differed in accuracy but received equal weights. Weights learned from training participants, and tested with a missing sensor, may help. These changes were not made here. No results were adjusted after viewing the test scores.

5 Conclusion

Combining acceleration from three body locations produced the strongest pooled result, reaching a macro F1 of 96.91%, and remained the most reliable choice across participants and activities. Adding gyroscopes through feature or decision fusion did not improve this pooled score, reducing macro F1 by 0.40 and 0.47 percentage points respectively, even though both fusion methods still outperformed every individual-location model. Participant-level bootstrap intervals showed this gap was not clearly resolved, and fusion helped several individual participants and activities while harming others, particularly around running-versus-jogging and sitting-versus-standing confusions. Feature-importance analysis confirmed the accelerometers drove most of the predictive signal, with gyroscopes contributing only about eleven percent. Decision fusion also cost roughly twice the training time of the baseline. Within this experiment, the simpler combined-accelerometer model remained the strongest and most efficient candidate; broader claims of gyroscope value would require external validation, continuous-stream testing, and a larger, more diverse participant sample.

References

Aguileta, A.A., Brena, R.F., Mayora, O., Molino-Minero-Re, E. and Trejo, L.A. (2019) 'Multi-sensor fusion for activity recognition—a survey', *Sensors*, 19(17), article 3808. https://doi.org/10.3390/s19173808.

Bian, S., Liu, M., Zhou, B. and Lukowicz, P. (2022) 'The state-of-the-art sensing techniques in human activity recognition: A survey', *Sensors*, 22(12), article 4596. https://doi.org/10.3390/s22124596.

Bragança, H., Colonna, J.G., Oliveira, H.A.B.F. and Souto, E. (2022) 'How validation methodology influences human activity recognition mobile systems', *Sensors*, 22(6), article 2360. https://doi.org/10.3390/s22062360.

Kapoor, S. and Narayanan, A. (2023) 'Leakage and the reproducibility crisis in machine-learning-based science', *Patterns*, 4(9), article 100804. https://doi.org/10.1016/j.patter.2023.100804.

Rehman, S.U., Ali, A., Khan, A.M. and Okpala, C. (2024) 'Human activity recognition: A comparative study of validation methods and impact of feature extraction in wearable sensors', *Algorithms*, 17(12), article 556. https://doi.org/10.3390/a17120556.

Sui, A., Sui, W., Liu, S. and Rhodes, R. (2023) 'Ethical considerations for the use of consumer wearables in health research', *Digital Health*, 9, article 20552076231153740. https://doi.org/10.1177/20552076231153740.

Thakur, D. and Biswas, S. (2023) 'Guided regularized random forest feature selection for smartphone-based human activity recognition', *Journal of Ambient Intelligence and Humanized Computing*, 14, pp. 9767–9779. https://doi.org/10.1007/s12652-022-03862-5.

Uddin, M.Z. and Soylu, A. (2021) 'Human activity recognition using wearable sensors, discriminant analysis, and long short-term memory-based neural structured learning', *Scientific Reports*, 11, article 16455. https://doi.org/10.1038/s41598-021-95947-y.